

Comparison of sex hormonal and metabolic profiles between omnivores and vegetarians in pre- and post-menopausal women.



The purpose of the present study was to investigate the sex hormonal and metabolic profiles in vegetarians and compare these with the profiles in omnivores. The design of the present study was cross-sectional. The study sample of pre- and post-menopausal women included forty-one omnivores and twenty-one vegetarians. Thereafter we determined: (1) plasma sex hormones, (2) fasting insulin, NEFA as well as apo-A and apo-B, (3) BMI, (4) a dietary profile (3 d dietary records), (5) physical activity and (6) total faecal excretion per 72 h and total urinary excretion per 72 h. Vegetarians showed higher levels of sex hormone-binding globulin (SHBG), apo-A, total faecal excretion per 72 h and total fibre intake as well as lower levels of apo-B, free oestradiol, free testosterone, dehydroepiandrosterone sulfate (DHEA-s) and BMI. Interestingly, after controlling for BMI, significant differences between groups still persisted except for apo-B. Moreover, stepwise regression analysis showed that total fibre intake explained 15.2 % of the variation in SHBG in our cohort, which accounted for the greatest source of unique variance. Results of the present study indicate that pre- and post-menopausal vegetarians present higher concentrations of SHBG, which could be explained, in part, by higher levels of fibre intake. This may explain, at least in part, the lower risk of developing type 2 diabetes.